KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 23 · MATH

The Tide Pool Survey

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Measuring in Eighths

Vikram is helping with a tide pool survey at Haystack Rock in Cannon Beach, Oregon. Warm up before the lesson, and show your work. Some problems may feel new, and that is okay!

1. Review (Week 22): Convert each measurement. a) 2 1/3 yards = _____ feet b) 2 gallons 3 quarts = _____ quarts c) 3.2 kilograms = _____ grams

2. Circle every fraction that is equal to 1/2.

   2/4 3/8 4/8 5/8 3/6

3. Add. Show how you rename the fractions first. a) 3/8 + 1/4 b) 1/2 + 3/8

4. Draw a number line from 0 to 1 with a tick mark every 1/8. Label 1/4, 1/2, and 3/4 on it.

5. Vikram measured 6 limpets: 1/2, 3/4, 1/2, 5/8, 3/4, and 1/2 inch. Which length shows up most often, and how many times?

KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 23 · MATH

The Tide Pool Survey

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Challenge: Survey Day

Show your work, rename with a common denominator, and label every answer with its unit.

1. Make a line plot in eighths of Uzoma's 8 anemone widths, in inches: 3/8, 1/2, 3/4, 4/8, 5/8, 6/8, 1/2, 7/8. Give it a title and label the units.

2. Use your line plot. a) How many anemones are wider than 1/2 inch? b) What is the difference between the widest and the narrowest anemone?

3. Two steps: six sample jars hold 1/4, 3/8, 1/2, 3/4, 7/8, and 1 cup of seawater. Find the total. If the water were shared equally, how much would each jar hold?

4. Find the mistake: Uzoma says 6 cups shared equally by 8 jars is 1 1/3 cups each, because 8 ÷ 6 = 1 1/3. In two or three complete sentences, explain his mistake and give the correct fair share.

★ **Challenge:** A line plot shows 4 X's at 5/8 inch, 3 X's at 3/4 inch, and 1 X at 1 1/4 inches. Find the total of all 8 measurements, then the fair share.

Looking Ahead: Week 24: Classifying two-dimensional figures. Sort triangles by sides and angles, climb the quadrilateral family tree, and find parallel sides and lines of symmetry in Pittsburgh bridge designs.

KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 23 · READING

Butterflies in Music City

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Words That Paint Pictures

Today you will visit a summer fiddle camp in Nashville, Tennessee, and read a story and a poem full of colorful language. Show what you already know. It is fine to guess!

1. Review (Week 22): Which is a **firsthand** account of Apollo 11? Explain how you know. a) Buzz Aldrin's words, radioed from the Moon b) a website article written in 2020

2. Is this sentence literal (exactly true) or figurative (paints a picture)? "The practice room was a refrigerator." What does it really mean?

3. Which sentence uses the word **like** or **as** to compare two things?

   The banjo twanged loudly. | Her voice was as sweet as honey. | The concert starts at seven.

4. People say a nervous person has "butterflies in their stomach." Write another saying you know, and tell what it really means.

5. Write one sentence about a time you felt nervous. Make your reader feel it by comparing your feeling to something else.

KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 23 · READING

Butterflies in Music City

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Figurative Language Explainer

Write in complete sentences with correct capitals, commas, and quotation marks.

1. Name each kind (simile, metaphor, personification, hyperbole, or idiom) and what it really means. a) The tuba grumbled in the corner. b) My banjo case weighs a million pounds.

2. Find the mistake. Alvin wrote: "'Her fingers flew like sparrows' is a metaphor." What is wrong? Fix his explanation.

3. Be the editor. Rewrite each sentence correctly. a) In the story Butterflies in Music City, Adisa plays a solo. b) "Take a deep breath" Mr. Aldo said.

4. Writing: In 4 or 5 sentences, explain how the figurative language in "Butterflies in Music City" shows Adisa's feelings changing. Quote one line correctly, and begin one sentence with However.

★ **Challenge:** Describe a thunderstorm at camp using a simile, a metaphor, and personification.

Looking Ahead: Week 24: Track how a character changes, and make inferences from text clues, in Sitka, Alaska.

KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 23 · SCIENCE

Decomposers and the Cycle of Matter

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: What Do I Already Know?

Answer with what you know right now. It is OK to guess!

1. **Review (Week 22):** A frog sees a fly and flicks out its tongue. Write the steps from stimulus to response, using the words receptors, neurons, brain and muscles.

2. What happens to a pumpkin left outside for a few months? Describe what you think you would see.

3. Is a mushroom a plant? Explain why or why not.

4. Draw a fallen leaf on the forest floor. Draw and label two living things you think might help break it down.

5. Predict: Where do you think the matter in a rotting log goes? List as many places as you can.

KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 23 · SCIENCE

Decomposers and the Cycle of Matter

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Follow the Matter

Use evidence from today's lesson. Write in complete sentences.

1. **Trace the matter.** A squirrel eats an acorn and later leaves droppings on the ground. Trace the matter from the air to the oak, the squirrel, the decomposers and back.

2. **Fix the compost problem.** Calixto's pile is dry and stays cool. What two things should he add, and how does each help the decomposers?

3. **Two-step problem (EPA, 2018).** Paper was 23.1 percent of U.S. waste and yard trimmings 12.1 percent. What percent were they together? How much more paper was there?

4. **Find the mistake.** "A fungus is a plant, so it makes food from sunlight." Explain what is wrong and how a fungus gets food.

★ **Challenge:** Draw the cycle of matter with four labeled stops: plants, animals, decomposers, air and soil. Add an arrow for energy from the Sun and explain why energy does not cycle.

Looking Ahead: Next week: Earth's Moving Plates: Earthquakes, Volcanoes and Mountains. Question to bring: why do earthquakes happen in some places more than others?

KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 23 · SOCIAL STUDIES

The Great Depression and the New Deal (1929–1941): Who Got Help?

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: What Do I Already Know?

Write what you know right now. Guesses are fine; this is your starting point.

1. **Review (Week 22):** Name one writer or musician of the Harlem Renaissance (for example, Langston Hughes, Zora Neale Hurston, Duke Ellington or Louis Armstrong). What did that person create?

2. What do you think a "depression" is when people talk about money and jobs?

3. Prediction: If one out of every four workers lost their job, what problems might families face? Name two.

4. How could a government help people during hard times? Give one idea, and explain who it would help.

5. Write one question you have about the 1930s.

KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 23 · SOCIAL STUDIES

The Great Depression and the New Deal (1929–1941): Who Got Help?

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Who Got Help, and Who Was Left Out?

Use evidence from today's lesson. Write in complete sentences.

1. **Two causes, one disaster.** Explain two causes of the Dust Bowl: one human choice and one natural cause.

2. **Do the math.** Suppose a Works Progress Administration worker earned $40 in a month and spent $28 on rent and food. How much was left each month? How much would that be after 3 months? Show your work.

3. **Find the mistake.** "The Social Security Act of 1935 helped every worker in America." Rewrite it so it is true, and name one group of workers it left out.

4. **Compare two sources.** FDR said in 1933, "This great Nation will endure." Dorothea Lange's 1936 caption says, "Destitute pea pickers in California." How do these two sources show different sides of the Depression?

★ **Challenge:** Write a short letter (3 or 4 sentences) that Mary McLeod Bethune might send to President Roosevelt asking that a New Deal program include Black Americans fairly. Use correct letter punctuation (greeting, commas and closing).

Looking Ahead: Next week: World War II at home and abroad (1941–1945). We meet the Navajo Code Talkers, the Tuskegee Airmen and women war workers, and learn about the Japanese Americans forced from their homes and the Holocaust.